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Customizable fitted apparatus

Abstract

A customizable fitted device for immobilizing injuries and a method for constructing the fitted device is provided. The fitted device can be used to stabilize an arm, wrist, hand, leg, knee, ankle, foot or other body parts through custom formation. The device can be formed by a sidewall having one or more sidewall sections that can be secured together using retaining clips positioned within retaining clip slots on the device. The device can further include one or more openings defined in the sidewall of each device section while maintaining rigidity in the sidewall. The device can be constructed by of creating a 3D image scan of the particular body part on which the device will be applied, creating a design of the device to precisely match the contours of the 3D image scan, and using a 3D printer to construct the device according to the design.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to currently pending U.S. patent application Ser. No. 16/677,434, filed on Nov. 7, 2019 to Diana Hall, entitled "Customizable Fitted Apparatus"; which in turn claimed priority to U.S. patent application Ser. No. 14/852,516 filed on Sep. 12, 2015 to Diana Hall, entitled "Customizable Fitted Apparatus"; which in turn claimed priority to U.S. patent application Ser. No. 62/162,743 filed on May 17, 2015 to Diana Hall, entitled "Amphibian Skin 3D Printed Clips and Cable Tie Slots"; and which also claimed priority to U.S. patent application Ser. No. 62/050,067, filed on Sep. 12, 2014 to Diana Hall, entitled "Amphibian Skin Custom-fit Exoskeleton Support Devices." The entire disclosures of each of these priority applications is hereby incorporated by reference.

BACKGROUND

1. Field of the Disclosure

(1) The present invention relates to devices, such as casts, splints, braces, and similar devices for immobilizing and stabilizing various portions of a user's body, including arms, elbows, wrists, hands, legs, knees, ankles, feet and other structures. Such devices are also used in veterinary applications and other various alternative uses.

2. Description of Related Art

(2) When a user's body portion needs stabilization, a cast, brace or similar device is commonly applied to immobilize the particular location of the identified body part. The cast or splint is commonly applied to an arm, leg, or joint, such as a knee, ankle, elbow or wrist. One common cast is a molded plaster or fiberglass cast formed around the location of the body part. A layer of padding, such as a cotton wrap, is applied around the body portion targeted and the plaster or fiberglass cast component is then applied and molded to conform to the patient's body. The cast is a single-use and non-removable device. While this style of casts is generally effective at immobilizing the targeted body portion and is generally inexpensive to produce, it has several limitations. One main limitation is that the cast is not removable. Thus, in order to clean the targeted body portion or subsequently examine the targeted body portion, the cast must be removed and a new cast applied. Another limitation is that this type of cast is non-breathable and non-waterproof. This can lead to skin irritation and other issues as well as damage to the cast when the patient encounters wet conditions.

(3) Accordingly, a need exists for a moldable cast or similar device that is effective at immobilizing a particular body part while being removable, reusable, lightweight, breathable, waterproof and structurally stable in multiple directions.

SUMMARY OF THE INVENTION

(4) The present invention is directed generally to a customizable fitted device for immobilizing

particular portions of a user's body. The device can be used to protect, stabilize, and/or immobilize the targeted body portion. According to one embodiment of the present invention, the device can include a generally enclosed sidewall divided into one or more shell-sections. The shell-section(s) can be joined together and form one or more longitudinal seams within the device sidewall. Each longitudinal seam can extend from the upper and lower terminal ends of the device. In an embodiment where there is only a single shell-section, the device contains sufficient flexibility, at least in a portion of the device, to allow the sidewall to be separated along the longitudinal seam and be removed from the user without damaging or destroying the device. In an embodiment where there are multiple shell-sections, the multiple longitudinal seams enable the device to be separated and removed from the user without damaging or destroying the device, even while the sidewall is substantially rigid.

(5) According to one embodiment, the enclosed sidewall can have a plurality of voids defined through the sidewall. The plurality of voids can define structural ribs in the sidewall that maintain structural rigidity while still permitting several openings within the sidewall. The device can also include primary openings at the upper and lower ends of the sidewall so that the device can be applied to only a portion of the user's body, such as the lower arm or leg with other portions of the user's body protruding from the device. Additionally, the device can include secondary openings defined within the sidewall to accommodate thumbs, finger, or other body portion.

(6) The device can further include one or more connector receivers positioned on the longitudinal seams of the device. The connector receivers can be used to hold the one or more sidewall sections together along the one or more longitudinal seams. Each connector receiver is positioned on both portions of the sidewall adjacent to the longitudinal seam. For example, in an embodiment where there are two sidewall- or shell-sections, each connector receiver is positioned on both sidewall sections adjacent to a longitudinal seam. The connector receiver can further be divided along the longitudinal seam so that one portion of the receiver remains on one side sidewall and the other portion of the receiver remains on the adjacent side of the sidewall when the sidewall is separated along the longitudinal seam. Each connector receiver also can include an opening or slot defined through the receiver. The opening can be configured to allow a connector or clip to be inserted through and hold the two portions of the connector receiver together.

(7) When the device is ready for use, the one or more sidewall sections can be placed around the target area of the user so that the one or more sidewall sections meet at the one or more longitudinal seams, thereby forming an enclosed sidewall around the target area. The connectors or clips can then be inserted through the receiver openings and secured to hold the connector receivers together. Once all connector receivers are secured, the device forms a rigid or supporting device around the target area. The device can then be easily removed from the user by removing the connectors or clips and separating the sidewall section(s).

(8) Other aspects and advantages of the present invention will be apparent from the following detailed description of the preferred embodiments and the accompanying drawing figures

(9) Other systems, methods, features and advantages of the disclosure will be, or will become, apparent to one of ordinary skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description and this summary, be within the scope of the disclosure, and be protected by the following claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale, emphasis instead being

placed upon illustrating the principles of the disclosure. Moreover, in the figures, like reference numerals designate corresponding parts throughout the different views.

(2) In the accompanying drawing, which forms a part of the specification and is to be read in conjunction therewith in which like reference numerals are used to indicate like or similar parts in the various views:

(3) FIG. 1 is a front perspective view of a fitted device for immobilizing a lower arm of a user in accordance with one embodiment of the present invention;

(4) FIG. 2 is a side perspective view of the fitted device of FIG. 1;

(5) FIG. 3 is a rear perspective view of the fitted device of FIG. 1;

(6) FIG. 4 is a side perspective view of the fitted device of FIG. 1;

(7) FIG. 5 is an exploded perspective view of the fitted device of FIG. 1, illustrating a first section and a second section of the fitted device separated prior to being applied to a user's arm in accordance with one embodiment of the present invention;

(8) FIG. 6 is a front perspective view of the fitted device of FIG. 1, illustrating the fitted device applied to a user's arm in accordance with one embodiment of the present invention;

(9) FIG. 7 is a front perspective view of a fitted device for immobilizing a hand of a user in accordance with one embodiment of the present invention;

(10) FIG. 8 is a front perspective view of the fitted device of FIG. 7;

(11) FIG. 9 is a rear perspective view of the fitted device of FIG. 7;

(12) FIG. 10 is a side perspective view of the fitted device of FIG. 7;

(13) FIG. 11 is an exploded perspective view of the fitted device of FIG. 7, illustrating a first section and a second section of the fitted device separated prior to being applied to a user's hand in accordance with one embodiment of the present invention;

(14) FIG. 12 is a rear perspective view of the fitted device of FIG. 7, illustrating the fitted device applied to a user's hand in accordance with one embodiment of the present invention;

(15) FIG. 13 is a top perspective view of a connector used to secure two sections of a fitted device in accordance with one embodiment of the present invention;

(16) FIG. 14 is a perspective view of a connector and fitted device for immobilizing a user's hand in accordance with one embodiment of the present invention, illustrating the connector inserted into a receiver slot and holding two sections of the fitted device together;

(17) FIG. 15 is a section plan view of the fitted device of FIG. 3, illustrating a connector inserted into a receiver slot for holding a first section and a second section of the fitted device together; and

(18) FIG. 16 is a perspective view of a connector and fitted device fitted device for immobilizing a user's hand in accordance with one embodiment of the present invention, illustrating the connector inserted into a receiver slot and holding two end of the fitted device together.

DETAILED DESCRIPTION

(19) The invention will now be described with reference to the drawing figures, in which like reference numerals refer to like parts throughout. For purposes of clarity in illustrating the characteristics of the present invention, proportional relationships of the elements have not necessarily been maintained in the drawing figures.

(20) The following detailed description of the invention references specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The present invention is defined by the appended claims and the description is, therefore, not to be taken in a limiting sense and shall not limit the scope of equivalents to which such claims are entitled.

(21) The present invention is directed toward a fitted device 10. FIG. 1 illustrates one embodiment of the present invention, where device 10 is configured for application to a user's arm. FIG. 7 illustrates another embodiment of the present invention where device 10 is configured for

application to a user's hand. As illustrated in the various figures, device **10** can be configured for application to an arm or hand of a human user; however, device **10** can also be suitable for application to other limbs or body parts, such as a leg or foot. Additionally, device **10** as described herein may be used for animals of all types and sizes.

(22) Device **10** can be used to support, stabilize and/or immobilize a limb or similar target area of a user by forming a partially enclosed structure around the target area. Device **10** can be formed as a substantially rigid structure without noticeable flexibility or device **10** can be formed with semi-flexible properties in all or some of device **10** depending on the particular intended application of device **10**. Device **10** can also substantially conform to the shape and contours of the target area by forming device **10** in accordance with a method of construction as described in greater detail below. As also described below, device **10** can be comprised of one or more sections that can enable device **10** to be easily applied, removed and re-applied to a user without damage to device **10**.

(23) As shown throughout the various figures, device **10** can comprise a generally enclosed sidewall **12** having a plurality of voids **14** defined therein. Voids **14** can have any number of different sizes and shapes and can be configured into particular patterns or non-uniformly entangled formations across sidewall **12**. As shown in FIG. 1, and in accordance with one embodiment of the present invention, voids **14** can be positioned in a certain configuration within sidewall **12** so as to form a plurality of structural ribs **16** between voids **14**. This configuration of voids **14** and ribs **16** can reduce the amount of material necessary to form device **10** and maintain rigidity of device **10** while providing the device with lightweight properties and enabling ventilation and breathability of device **10** with regard to the user's limb or other target area where device **10** is applied while conforming to manufacturing requirements.

(24) Device **10** can further comprise one or more shell-sections **18**, which can be joined together to form the enclosed sidewall **12**. As shown in FIG. 16, device **10** comprises a shell-section **18**, which can be joined at its ends **26** and **28** to form the enclosed sidewall **12**. As shown in FIGS. 5 and 11, device **10** can comprise a shell-section **18** and a shell-section **20**, which can be joined together to form the enclosed sidewall **12**. Device **10** can also comprise more than two sidewall sections in alternative embodiments of the present invention (this alternative embodiment is not shown in the figures).

(25) In an embodiment where device **10** includes one shell-section **18**, sidewall **12** can include a sidewall seam **22** defined longitudinally through sidewall **12**. Sidewall seam **22** can define a first and a second longitudinal ends **26** and **28** of shell-section **18**, which join together to form sidewall seam **22**. Sidewall seam **22** can be configured in a general wave-like configuration or a generally straight configuration. Alternative configurations can also be used, especially configurations that further the structural stability of the device **10**. A wave-like configuration can also be configured to provide structural stability of the device **10**.

(26) As best illustrated in FIGS. 1 and 3, in an embodiment where device **10** includes shell-sections **18** and **20**, shell-sections **18** and **20** can be defined by sidewall seams **22** and **24** defined longitudinally through sidewall **12** and thereby separating sidewall **12** into shell-sections **18** and **20**. As shown in FIGS. 1 and 3, when both shell-sections **18** and **20** form sidewall **12**, shell-section **18** has first and second longitudinal edges **26** and **28** that correspond to first and second longitudinal edges **30** and **32** of shell-section **20**. Longitudinal edges **26** and **30** combine to form first seam **22** while longitudinal edges **28** and **32** combine to form second seam **24**. As shown in FIG. 3, seams **22** and **24** (along with corresponding longitudinal ridges **26-32**) can be configured in a general wave-like pattern, as shown in FIG. 3, or can have a straighter configuration as shown in FIG. 7. Other configurations or orientations can also be used for seams **22** and **24** so long as seams **22** and **24** are suitable for joining shell-sections **18** and **20** together. Seams **22** and **24** can be formed (and correspondingly, enclosed sidewall **12**) when first longitudinal edge **26** of shell-section **18** is joined to conforming first longitudinal edge **30** of shell-section **20** and second longitudinal edge **28** of shell-section **18** is joined to conforming second longitudinal edge **32** of shell-section **20**. When

shell-sections **18** and **20** are joined together at seams **22** and **24** to form enclosed sidewall **12**, adjacent longitudinal edges **26** and **30** and **28** and **32** can be positioned with minimal space defined between.

(27) Enclosed sidewall **12** can also define primary end openings **34** and **36** of device **10** at sidewall's **12** terminal or axial ends, as best illustrated in FIGS. **4** and **14**. Primary end openings **34** and **36** can accommodate a user's arm, hand, fingers, leg, foot, etc. extending away from device **10** in either direction. Depending on the particular application or use, device **10** can also include secondary openings **38** for thumbs, fingers or similar extremities. As shown in FIG. **2**, device **10** can include a secondary opening **38** for receiving a user's thumb, according to one embodiment of the present invention.

(28) Device **10** can also include one or more connector receiver pairs **40** located on seams **22** and/or **24** and comprising two connector receivers **42**. As described below, each connector pair **40** can be separable into two portions to allow shell-section **18** to be separated at its ends **26** and **28** or to be separated from shell-section **20**. FIG. **2** illustrates one embodiment of device **10** having two receiver pairs **40** along each seam **22** and **24** while FIGS. **9** and **10** illustrate another embodiment of device **10** having only one receiver pair **40** along each seam **22** and **24**. FIG. **16** illustrates yet another embodiment of device **10** having only one receiver pair along seam **22**, which connects the first and second longitudinal ends **26** and **28** of shell-section **18** together. Any number of additional combinations of receiver pairs **40** may be suitable depending on the particular configuration and shape of device **10**. Receiver pairs **40** can be used in conjunction with connectors **44** to secure and hold shell-sections **18** and **20** together along seams **22** and **24**, as best shown in FIG. **14**, or to secure shell-section **18** together at its ends **26** and **28**, as best shown in FIG. **16**.

(29) Each connector receiver **42** can extend axially away from sidewall **12** adjacent to seam **22** or **24** (and corresponding longitudinal edge **26-32**) as best shown in FIG. **11**. As also shown in FIG. **11**, each connector receiver **42** can also have first and second raised sidewalls **46** and **48** and a top wall **50** extending therebetween and defining the upper boundary of the receiver **40**. Defined within each connector receiver **40** can be a slot **52** extending from first raised sidewall **46** to second raised sidewall **48** for receiving a section of connector **44**.

(30) As shown in FIG. **11**, second raised sidewall **48** of each receiver **42** can conform to the corresponding longitudinal edge **26-32** adjacent to the receiver **42** so that each receiver **42** does not extend beyond seam **22** or **24** and first and second shell-sections **18** and **22** can be joined at seams **22** and **24** with a conforming fit. As best shown in FIGS. **1** and **3**, each connector receiver **42** on first longitudinal edge **26** of shell-section **18** corresponds to each connector receptacle **42** on first longitudinal edge **30** of shell-section **20** so that the receivers **42** align along seam **22** and form receiver pairs **40**. Similarly, each connector receiver **42** on second longitudinal edge **28** of shell-section **18** can align with each connector receiver **42** on second longitudinal edge **32** of section-shell **18** to form receiver pairs **40**. Alternatively, when sidewall **12** comprises only one shell-section **18**, as shown in FIG. **16**, each connector receiver **42** on first longitudinal edge **26** corresponds to a connector receiver **42** on second longitudinal edge **28** so that the receivers **42** align along seam **22** to form a receiver pair **40**. Pursuant to this alignment, slots **52** of corresponding connector receivers **42** in a receiver pair **40** align to form a continuous slot or opening **52** through the receiver pair **40** when shell-sections **18** and **20** are joined together, as best shown in FIG. **8**, or when ends **26** and **28** of shell-section **18** are joined together, as best shown in FIG. **16**.

(31) Connectors **44** can be used to lock receivers **42** in each connector receiver pair **40** together to join shell-section **18** at its ends as shown in FIG. **16**, or to join shell-section **18** and **20** together as shown in FIG. **14**. Connectors **44** can comprise many different types of fasteners depending on the particular embodiment of the present invention. FIG. **13** illustrates one specific embodiment of connector **44** in the form of a clip. As shown in FIG. **13**, clip **44** can comprise an upper arm **54** and a lower arm **56** connected together by a resilient bridge **58** at one end of each arm. At its end opposing bridge **58**, clip **44** can include a retaining rib **60** extending downward from upper arm **54**

into a groove **62** defined in lower arm **56**. Retaining rib **60** can include a lower edge **64** that has an inward-extending lip **66** as best shown in FIG. **13**. Lip **16** facilitates securing connector pairs **42** together as described in greater detail below. Resilient bridge **58** allows upper arm **54** (and attached rib **60**) to be flexed upward away from lower arm **56** by a user to fasten clip **44** to receptacle pair **40**, as also described in greater detail below.

(32) Lower arm **56** can be configured to be inserted into and through receiver slots **52** of a connector receiver pair **40** when joined together. According to this configuration, lower arm **56** preferably has a thickness T_c that is less than a height T_r of receiver slot **52** of receiver pair **40** as best shown in FIG. **15**. Lower arm **56** also preferably has a width Y_c that is less than a width Y_r of receiver slot **52** as best shown in FIG. **14**. As shown in FIG. **15**, the distance between resilient bridge **58** and retaining rib **60** is preferably slightly greater than the length of receiver pair **40** so that clip **44** tightly joins receptacles **42** together.

(33) Connectors **46** can also comprise any number of alternative embodiments that are suitable for maintaining a joined connected between first and second shell-halves **18** and **20**. In an alternative embodiment (not shown), receiver slots **52** of each receiver pair **40** include a divider wall (not shown) that allows a cord, string, wire or similar tie to semi-permanently secure receivers **42** of each receiver pair **40** together.

(34) As best shown in FIG. **5**, prior to use, shell-sections **18** and **20** can be separated from one another. Device **10** can then be used by positioning shell-sections **18** and **20** around the user's target area (such as an arm or leg) and joining first longitudinal edges **26** and **30** to form seam **22** and joining second longitudinal edges **28** and **32** to form seam **24**, as best illustrated in FIG. **6**. When shell-sections **18** and **20** are joined together along seams **22** and **24**, corresponding connector receivers **42** of each receiver pair **40** can be joined together. In an embodiment where sidewall **12** comprises only shell-section **18**, the longitudinal ends **26** and **28** can be separated along seam **22** until device **10** can be applied to the target area of the user, where longitudinal seams **26** and **28** can be joined back together.

(35) Connectors **44** can then be used to lock or secure shell-sections together as shown in FIGS. **14-16**. When a clip **44** is used as shown in FIGS. **14-16**, lower arm **56** of each clip **44** can be inserted into receiver slot **52** of the corresponding receiver pair **40** by raising upper arm **54** away from lower arm **56** until the distance between a lower edge **64** of retaining rib **60** and lower arm **56** is greater than the width top wall **50** of receivers **42**. Once lower arm **56** of clip **44** is fully inserted through slot **52**, retaining rib **60** snaps downward partially into groove **62** to lock clip **44** in place due to the resiliency of bridge **58**. When fully inserted, clip **44** restricts receivers **42** in a receiver pair **40** from separating because top walls **50** of each receiver **42** enclosed by upper arm **54**, lower arm **56**, resilient bridge **58** and retaining rib **60** of clip **44**. Lip **66** on retaining rib **64** can also be positioned partially within slot **52** and thereby prevent upper arm **54** and lower arm **56** from separating. After all clips **44** are inserted into each receiver pair **40** and enclosed around top walls **50**, device **10** can be rigidly secured and sidewall **12** can form around the user's arm or other target area.

(36) Device **10** can then be removed from the user by removing connectors **44** and separating first and second shell sections **18** and **20**. Because lip **66** can be partially inserted into slot **52**, a user can be required to push the resilient bridge laterally against the first sidewall **36** of connector receiver **42** in order to allow lip **66** to exit slot **52**. Connectors **44** can then be removed by lifting upper arm **56** away from lower arm **58** and sliding lower arm **58** out of slot **52**. Once all connectors **44** are removed from receiver pairs **40**, shell-sections **18** and **20** can be removed from the user. Alternatively, when sidewall **12** includes one shell-section **18**, longitudinal ends **26** and **28** can be separated in or to allow the device **10** to be removed from the user. Device **10** can then be reapplied in the manner described above.

(37) Collectively, connectors **44** and receiver pairs **40** provide structural stability for the device **10**. Each connector **44** can securely hold the connector receivers **44** within a connector receiver pair **40**

together and restrict movement along several different directional axes. The sizing of the upper arm **54** and lower arm **56** thicknesses in relation to the top wall **50** thicknesses of the connector receivers **42** restricts movement of one longitudinal end **26-32** relative to another longitudinal end **26-32** along sidewall seam **22** or **24** in a first direction. Lip **66** on retaining rib **64** also assists in preventing this direction of movement by preventing upper and lower arms **54** and **56** from separating unless lip **66** is removed from slot **52**. The length of clip **44** between resilient bridge **58** and retaining rib **60** relative to the length of a connector receiver pair **40** restricts movement of a paired longitudinal ends **26-32** relative to one another along seam **22** or **24** in a second direction. Resilient bridge **58** and retaining rib **60** also restrict movement of paired longitudinal ends **26-32** relative to one another along seam **22** or **24** in a third direction. As a result, device **10** can maintain structural rigidity about sidewall **12** when connectors **44** are secured onto clip receiver pairs **40**.

(38) Device **10** can be sized, constructed, and designed according to a number of different methods or processes. According to one process, different information can be collected about the intended user of the device. Such information can be collected by various types of equipment, such as scanners and other similar devices. Some or all of this information can also be collected manually. The collected information can then be used to design device **10** according to specific parameters. The information can be used in conjunction with one or more computer analysis programs or software, and other technology in order to develop the design of device **10**. The information can also be in a human design process to develop device **10**.

(39) The design of device **10** can take into account several different parameters. For example, the design can be dependent on the user, the intended application of device **10**, desired immobilization, desired stability, size, shape, temperature, conditions, etc., each of which can affect the material used, the type of scanning device or printer used, and the like. According to one process, the information collected is used to design a device **10** specifically customized to a user with lightweight, breathable, waterproof, and structural stability properties. Further, the process can incorporate additional factors, such as the degree of removability, degree of immobilization of the target area, among others. Based on the foregoing, the design process, according to one embodiment of the present invention, can collect different information specific to a user, combine the collected information with additional design information non-specific to the user, combine, manipulate, and factor the collected and additional information using computer implemented devices, a human design process or a combination of the two in order to develop the design of device **10** specifically adapted to the user in structure and functionality.

(40) According to one method, device **10** can be formed using a 3D scanner to precisely identify the shape and contours of the user's target area. The 3D scanner can incorporate a hand-held, white-light reflective scanner that can be selectively positioned around the user's arm or other body part in which device **10** will be applied to create a 3D image of the target area. Additional information can be collected and incorporated in certain applications. 3D CAD software, along with other design software, can then be used to design the device **10** to precisely fit around 3D image by effectively drawing the shape of the device **10** onto the 3D image model. Voids **14** and corresponding structural ribs **16** can then be selectively designed into device **10**. Seams **22** and/or **24** can then be provided onto the design of device **10** to form shell-section **18** and shell-section **20**, if applicable. Connector receiver pairs **40** can then be provided onto device **10** along seams **22** and/or **24**. Connectors **44** can also be designed to correspond to receiver pairs **40**. Device **10** can then be constructed by printing each shell-section **18** and **20** (if applicable) using a 3D printer. Device **10** can be constructed from several different materials depending on the particular application of device **10**. Device **10** can be constructed from a uniform material or a combination of materials. The specific material used can affect the flexibility, rigidity, structural stability, weight and other properties of the device. In one embodiment, device **10** is constructed from a rigid material such as rigid plastic. In an alternative embodiment, device **10** is constructed from a semi-rigid or semi-flexible material, such as flexible plastic or woven fabric material.

(41) From the foregoing, it will be seen that this invention is one well adapted to attain all the ends and objects hereinabove set forth together with other advantages which are inherent to the structure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is within the scope of the claims. Since many possible embodiments of the invention may be made without departing from the scope thereof, it is also to be understood that all matters herein set forth or shown in the accompanying drawings are to be interpreted as illustrative and not limiting.

(42) The constructions described above and illustrated in the drawings are presented by way of example only and are not intended to limit the concepts and principles of the present invention. Thus, there has been shown and described several embodiments of a novel invention. As is evident from the foregoing description, certain aspects of the present invention are not limited by the particular details of the examples illustrated herein, and it is therefore contemplated that other modifications and applications, or equivalents thereof, will occur to those skilled in the art. The terms “having” and “including” and similar terms as used in the foregoing specification are used in the sense of “optional” or “may include” and not as “required”. Many changes, modifications, variations and other uses and applications of the present construction will, however, become apparent to those skilled in the art after considering the specification and the accompanying drawings. All such changes, modifications, variations and other uses and applications which do not depart from the spirit and scope of the invention are deemed to be covered by the invention which is limited only by the claims which follow.

Claims

1. A customized removable fitted immobilization device for a body portion comprising: a sidewall formed by a first shell section and a second shell section, each of the first and the second shell sections having a first longitudinal edge and a second longitudinal edge, wherein the first longitudinal edge and the second longitudinal edge of each shell section collectively forms a plurality of longitudinal edges, and wherein the sidewall is configured to conform to a shape and one or more contours of the body portion of a specific user as manufactured and prior to being applied to the body portion of the specific user; at least one longitudinal sidewall seam extending longitudinally along the sidewall between an upper sidewall edge and a lower sidewall edge, wherein each longitudinal sidewall seam removably joins two longitudinal edges of the plurality of longitudinal edges such that the two longitudinal edges making up the longitudinal side wall seam are directly contiguous to each other along a full length of each longitudinal edge from a first end of the device to a second end of the device; wherein the first longitudinal edge and the second longitudinal edge on each of the first shell section and the second shell section extend perpendicular to an outer surface of the sidewall; wherein the plurality of longitudinal edges of the first shell section are configured so as to conform to the plurality of longitudinal edges of the second shell section so as to join the first shell section to the second shell section; a first connector receiver pair positioned along a first at least one longitudinal sidewall seam, the first connector receiver pair comprising: a first connector receiver positioned on the sidewall and adjacent to the first at least one longitudinal sidewall seam, the first connector receiver having a slot defined therein; and a second connector receiver positioned on the sidewall and adjacent to the first at least one longitudinal sidewall seam and positioned opposing the first connector receiver relative to the first at least one longitudinal sidewall seam, the second connector receiver having a slot defined therein; and a first connector configured for securing the first and second connector receivers of the first connector pair together when the sidewall is joined along the first at least one longitudinal sidewall seam; wherein, when the first connector secures the first and second connector receivers, the first connector receiver pair and the first connector together provide structural stability for the device in three directional axes; wherein the sidewall is substantially rigid without noticeable

flexibility.

2. The device of claim 1, wherein the first at least one longitudinal sidewall seam has a configuration such that the two longitudinal edges making up the longitudinal sidewall seam conform to one another other, such that the longitudinal sidewall seam provides structural stability to the device.
3. The device of claim 1, wherein the first at least one longitudinal sidewall seam has a wave-like configuration.
4. The device of claim 1, wherein the first shell and the second shell are configured such that the device can be easily applied, removed, and re-applied to the body portion of the specific user without damage to the device.
5. The device of claim 1, wherein the device is constructed by printing each of the first shell and the second shell using a 3D printer.
6. The device of claim 5, wherein the sidewall has a plurality of voids defined in the sidewall, and a configuration of the plurality of voids defines a plurality of structural ribs, each structural rib located between two or more voids of the plurality of voids.
7. The device of claim 6, wherein the plurality of voids is arranged in a configuration that provides structural stability for the device.
8. The device of claim 1, wherein the first connector receiver is positioned adjacent to the second connector receiver so that the slots of the first and the second connector receivers align to form a continuous opening of the connector receiver pair when the sidewall is joined together at the first at least one longitudinal sidewall seam.
9. The device of claim 8, wherein the first and second connector receivers each contain a receiver wall located along the first at least one longitudinal sidewall seam, and wherein the receiver walls have a configuration relative to one another that provides structural stability for the device along the first at least one longitudinal sidewall.
10. The device of claim 1, wherein each connector receiver further comprises: a top wall extending away from the device sidewall in a generally perpendicular direction; a first receiver wall positioned laterally inward from the first at least one longitudinal sidewall seam and extending generally perpendicular between the device sidewall and the receiver top wall; and a second receiver wall positioned along the first sidewall seam and extending between the device sidewall and the receiver top wall, the second receiver wall generally conforming to the first at least one longitudinal sidewall seam; wherein the receiver slot is defined between the first and the second receiver walls and extends generally perpendicularly from the first and the second lateral walls.
11. The device of claim 10, wherein the second receiver wall of the first connector receiver conforms to the second receiver wall of the second connector receiver.
12. The device of claim 1, wherein the connector comprises a configuration that structurally stabilizes the device in the three directional axes when the connector is securing the first connector receiver pair.
13. The device of claim 1, wherein the device includes a primary opening at each of the first end of the device and the second end of the device, and further includes at least one secondary opening.
14. The device of claim 1, wherein the at least one longitudinal sidewall seam comprises a first longitudinal sidewall seam and a second longitudinal sidewall seam, the first longitudinal sidewall seam removably joins a first longitudinal edge of the first shell section and a first longitudinal edge of the second shell section such that the first longitudinal edge of the first shell section and the first longitudinal edge of the second shell section are adjacent to each other, and the second longitudinal sidewall seams removably joins a second longitudinal edge of the first shell section and a second longitudinal edge of the second shell section such that the second longitudinal edge of the first shell section and the second longitudinal edge of the second shell section are adjacent to each other.
15. The device of claim 1, wherein the first connector receiver and the second connector receiver are integrally connected to the sidewall.

16. A customized removable fitted immobilization device for a body portion comprising: a first shell section and a second shell section, each of the first shell section and the second shell section having a first longitudinal edge and a second longitudinal edge; a sidewall formed by the first shell section and the second shell section, the sidewall being configured to conform to a shape and one or more contours of the body portion of a specific user as manufactured and prior to being applied to the body portion of the specific user; a first longitudinal sidewall seam and a second longitudinal sidewall seam; and wherein the first longitudinal sidewall seam removably joins the first longitudinal edge of the first shell section and the first longitudinal edge of the second shell section such that the first longitudinal edge of the first shell section and the first longitudinal edge of the second shell section are contiguous to each other along a full length of each longitudinal edge from a first end of the device to a second end of the device; wherein the second longitudinal sidewall seam removably joins the second longitudinal edge of the first shell section and the second longitudinal edge of the second shell section such that the second longitudinal edge of the first shell section and the second longitudinal edge of the second shell section are contiguous to each other along a full length of each longitudinal edge from a first end of the device to a second end of the device; wherein the first longitudinal edge and the second longitudinal edge on each of the first shell section and the second shell section extend perpendicular to an outer surface of the sidewall; wherein the plurality of longitudinal edges of the first shell section are configured so as to conform to the plurality of longitudinal edges of the second shell section so as to join the first shell section to the second shell section; and wherein the sidewall is substantially rigid without noticeable flexibility.

17. The device of claim 16, wherein the sidewall has a plurality of voids defined in the sidewall, and a configuration of the plurality of voids defines a plurality of structural ribs, each structural rib located between two or more voids of the plurality of voids; and wherein the plurality of voids is arranged in a configuration that provides structural stability for the device.

18. The device of claim 17, wherein the device includes a primary opening at each of the first end of the device and the second end of the device, and further includes at least one secondary opening.

19. A customized removable fitted immobilization device for a body portion comprising: a first shell section and a second shell section, each of the first shell section and the second shell section having a first longitudinal edge and a second longitudinal edge; a sidewall formed by the first shell section and the second shell section, the sidewall being configured to conform to a shape and one or more contours of the body portion of a specific user as manufactured and prior to being applied to the body portion of the specific user; a first longitudinal sidewall seam and a second longitudinal sidewall seam; and wherein the first longitudinal sidewall seam removably joins the first longitudinal edge of the first shell section and the first longitudinal edge of the second shell section such that the first longitudinal edge of the first shell section and the first longitudinal edge of the second shell section are contiguous to each other along a full length of each longitudinal edge from a first end of the device to a second end of the device; wherein the second longitudinal sidewall seam removably joins the second longitudinal edge of the first shell section and the second longitudinal edge of the second shell section such that the second longitudinal edge of the first shell section and the second longitudinal edge of the second shell section are contiguous to each other along a full length of each longitudinal edge from a first end of the device to a second end of the device; and wherein the sidewall is substantially rigid without noticeable flexibility.

20. The device of claim 19, further comprising: a first connector receiver pair positioned along one of the first longitudinal sidewall seam and the second longitudinal sidewall seam, the first connector receiver pair comprising: a first connector receiver positioned on the sidewall and adjacent to the longitudinal sidewall seam, the first connector receiver having a slot defined therein; and a second connector receiver positioned on the sidewall and adjacent to the longitudinal sidewall seam and positioned opposing the first connector receiver relative to the longitudinal sidewall seam, the second connector receiver having a slot defined therein.

